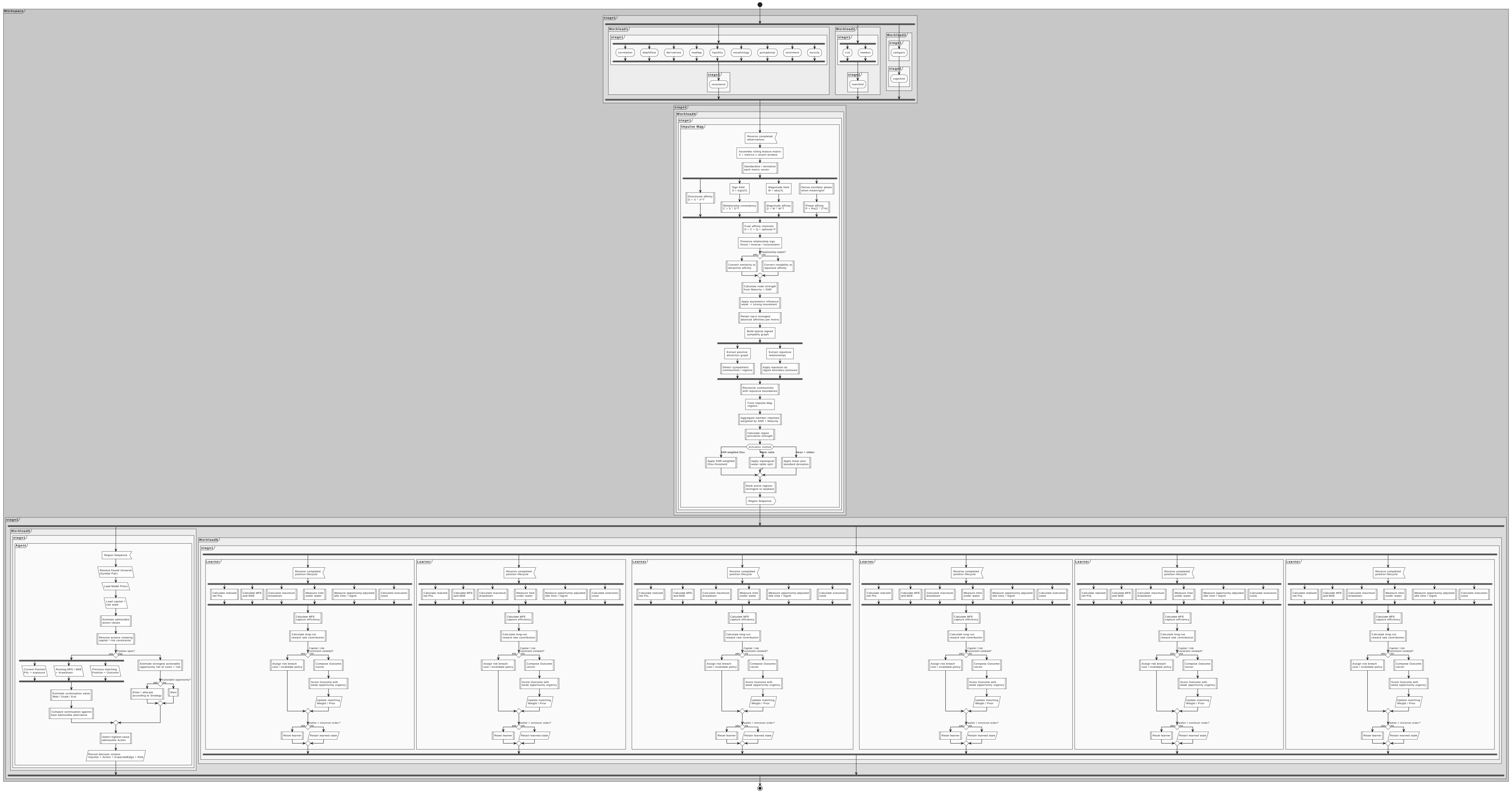
Workspace
stage1
Workload1
stage1
correlation
depthflow
derivatives
leadlag
liquidity
morphology
pumpdump
sentiment
toxicity
stage2
resonance
Workload2
stage1
cvd
hawkes
stage2
manifold
Workload3
stage1
category
stage2
cognition
stage2
Workload4
stage1
Impulse Map
Receive completed observations
Assemble rolling feature matrix X = metrics x recent window
Standardize / normalize each metric vector
Directional affinity D = X * X^T
Sign field S = sign(X)
Relationship consistency C = S * S^T
Magnitude field M = abs(X)
Magnitude affinity Q = M * M^T
Derive oscillator phase when meaningful
Phase affinity P = Re(Z * Z^H)
Fuse affinity channels D + C + Q + optional P
Preserve relationship sign Direct / Inverse / Inconsistent
Relationship stable?
yes
no
Convert similarity to attractive affinity
Convert instability to repulsive affinity
Calculate node strength from Maturity + SNR
Apply asymmetric influence weak -> strong movement
Retain top-k strongest absolute affinities per metric
Build sparse signed sympathy graph
Extract positive attraction graph
Detect sympathetic communities / regions
Extract repulsive relationships
Apply repulsion as region boundary pressure
Reconcile communities with repulsive boundaries
Form Impulse Map regions
Aggregate member impulses weighted by SNR + Maturity
Calculate region activation strength
Activation method
SNR-weighted Otsu
Water table
Mean + stddev
Apply SNR-weighted Otsu threshold
Apply topological water-table split
Apply mean plus standard deviation
Rank active regions strongest to weakest
Region Sequence
stage3
Workload5
stage1
Agent
Region Sequence
Resolve Keyed Universe (Symbol Pair)
Load Model Priors
Load capital + risk state
Estimate admissible action values
Remove actions violating capital / risk constraints
Position open?
yes
flat
Current Position PnL + exposure
Running MFE / MAE + drawdown
Previous matching Position + Outcome
Estimate continuation value Wait / Scale / Exit
Compare continuation against best admissible alternative
Estimate strongest actionable opportunity net of costs + risk
Actionable opportunity?
yes
no
Enter / allocate according to Strategy
Wait
Select highest-value admissible Action
Record decision context Impulse + Action + ExpectedEdge + Risk
Workload6
stage1
Learner
Receive completed position lifecycle
Calculate realized net PnL
Calculate MFE and MAE
Calculate maximum drawdown
Measure time under water
Measure opportunity-adjusted idle time / regret
Calculate execution costs
Calculate MFE capture efficiency
Calculate long-run reward rate contribution
Capital / risk constraint violated?
yes
no
Assign risk breach cost / invalidate policy
Compose Outcome vector
Score Outcome with weak opportunity urgency
Update matching Weight / Prior
wallet < minimum order?
yes
no
Reset learner
Retain learned state
Learner
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